Phase-Lock Nullification and the Wave-Mechanics Reduction of Time

In relativistic physics, time is treated as an independent background dimension (a fundamental property) along which events unfold. However, paradoxes such as infinite time dilation at gravitational event horizons ($r \to r_s$) and the irreversibility of thermodynamic entropy suggest that time may not be a fundamental independent metric.

This paper proposes the wave-mechanics reduction of time. By arguing that temporal intervals are dependent metrics arising entirely from active phase progression (change in frequency baseline) and cyclic frequency. In Section 1, examines why backwards time travel is forbidden by wave remodulation and entropy decoherence. In Section 3, explores relational cyclic degeneracy ($f = k$). In Section 4, provides a mathematical evaluation of phase-lock nullification, resolving the Schwarzschild limit as a coordinate breakdown rather than an infinite temporal stretch. Finally, Section 5 integrates quantum zero-point energy constraints to demonstrate why global time cannot completely freeze.

[Section 7 discusses limitations to this model, Section 8 gives list of definitions for all symbols, operators and equations used in this paper and Section 9 provides references used for this paper]

Introduction (Section 1)

Travelling back in time is simply unachievable because the past is a different configuration of frequency, amplitude and wavelength (f, A, λ). Once energy is transferred to remodulate the wave, that configuration will cease to exist.

According to thermodynamics, when energy is dissipated into macrostates of higher entropy, the open system's baseline frequency undergoes decoherence. Hence, recovering that exact baseline frequency will require restoring the momentum and baseline frequency of every quantum entity in the universe, which is beyond our reach and is practically not possible but never zero. Once energy transfers or system disruptions alter the baseline frequency of a system $(\phi \to \phi')$, the past state is overwritten rather than preserved in a parallel continuum.

Structural Resistance: From Solids to Vacuum Fields (Section 2)

Unless the configuration of the wave disappears, it can remain static (which is equivalent to stopping time), which happens when there are zero disturbances and no energy transfers leading to no phase remodulations; the metric of time vanishes. Since time is a measure of Frequency's cycles, it cannot be measured if waves are static and nothing is cycling.

For instance, an iron block has a localised internal frequency spectrum constrained by chemical bonds and subparticles only; a small, sharp force can easily disrupt its baseline frequency phase structures. (Malleability) The force shatters the solid (an energy transfer), dissipating heat and remodulating the material into a new physical configuration (an alternate baseline frequency). This doesn't take infinite force, as the system is factored to outside energy transfers.

However, to freeze the flow of time itself (an active propagation of frequency) is an exponentially larger model of the solid analogy. It isn't just one stream of frequency of a localised object but for an entire local vacuum field. To arrest its phase changes and force a static frequency stream where $\Delta f = 0$ (reaching a zero-change singularity), a small sharp force is insufficient. The force needed is scaled up to an infinite accumulation of mass-energy density- the singularity of a black hole. (where time stops or collapses)

Relational Degeneracy and Cyclic Repetition ($f = k$) (Section 3)

Even if the frequency is non-zero ($f = k$), **time-as-a-measurement still vanishes because there is no history of relative change.**Just as **Relational Quantum Mechanics** suggests that physical properties and values do not exist in objects as their own but exist relationally. A single wave cycling perfectly at $f = k$ with zero environmental disruptions or baseline frequency changes has no relational difference between its cycles. To the universe, that state is indistinguishable from stillness. Hence, duration collapses.

Let a wave system be sampled such that its frequency is constant ($f = k$), but every sequential cycle of the wave constitutes an identical phase copy with zero relative variation ($\Delta\theta_n = \Delta\theta_{n+1}$). Under this still frequency baseline condition, the relational history required to register temporal intervals is nullified. Although frequency remains mathematically non-zero, the absence of differential phase (change in frequency baseline) progression collapses duration into a dimensionless point, demonstrating that time requires informational state-change rather than mere cyclic repetition.

General Relativity, Redshift, and the Schwarzschild Limit (Section 4)

In General Relativity, as light climbs out of a deep gravitational fall, it suffers redshift. Redshift is not just an optical effect but also when the photon's wave crests are stretching further and further apart. In terms of pure wave mechanics, its observed frequency is dropping ($f \to 0$).

Calculating time dilation using the formula: where time dilation is a differential frequency suppression and phase-rate modulation.

$$dt_{obs} = \frac{dt_0}{\sqrt{1-\frac{r_s}{r}}}$$

As an infalling object reaches the Schwarzschild radius ($r = r_s$), the denominator goes to 0. Standard physics divides a non-zero interval by zero and claims:

$$Timedt = \frac{something}{0} = \infty$$

This leads to the bizarre conclusion that an infalling astronaut, to the point of deep gravity, hangs frozen at the horizon for infinite years from the perspective of an outside observer.

Time is not a background line that stretches. It is a dependent phase ratio:

$$dt = \frac{d\theta}{2\pi f}$$

When gravitational forces the local wave frequency to zero ($f \to 0$), **the numerator does not stay constant.** Phase change (baseline frequency remodulation) ($d\theta$) is driven *by* frequency. As frequency drops, phase accumulation slows down at the exact same rate:

$$\lim_{f \to 0} \frac{d\theta}{dt} = \lim_{f \to 0} (2\pi f) = 0$$

Because the speed of light (c) is constant, if you stretch the wavelength (λ) toward infinity, the frequency (f) must **collapse to zero**. Frequency is how many wave crests pass by per second; if the crests are stretched infinitely far apart, they stop arriving altogether.

Let the background spacetime be described by the standard Schwarzschild metric under standard spherical coordinates:

$$ds^2 = -\left(1 - \frac{r_s}{r}\right)c^2dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1}dr^2 + r^2d\Omega^2$$

In traditional General Relativity, as the radial coordinate approaches the event horizon ($r \to r_s$), the temporal component of the metric vanishes while the spatial component diverges toward infinity. However, by substituting our wave-mechanics definition of local time—where proper time interval dt is fundamentally coupled to local wave phase accumulation ϕ and frequency f via $dt = \frac{d\phi}{2\pi f}$—the local invariant interval transforms from a background coordinate divergence into a local phase-rate ratio:

$$ds^2 = -\left(1 - \frac{r_s}{r}\right)c^2\left(\frac{d\phi}{2\pi f}\right)^2 + \left(1 - \frac{r_s}{r}\right)^{-1}dr^2$$

As $r \to r_s$, the local frequency suppression ($f \to 0$) and phase-rate drop ($d\phi \to 0$ trajectory convergence) occur symmetrically. This converts the asymptotic infinity $\left(\frac{non-zero}{0}\right)$ into a bounded coordinate indeterminacy $\left(\frac{0}{0}\right)$, resolving the singularity not as an infinite stretch of time, but as a local manifold phase-lock nullification.

ALGEBRAIC BREAKDOWN:

In standard gravitational redshift, local wave frequency scales with the metric potential:

$$f(r) = f_0\sqrt{1 - \frac{r_s}{r}}$$

Squaring both sides gives:

$$f^2 = f_0^2\left(1 - \frac{r_s}{r}\right)$$

Focusing on the time-time component of your substituted metric:

$$ds_{time}^2 = -\left(1 - \frac{r_s}{r}\right)c^2\left(\frac{d\phi}{2\pi f}\right)^2$$

Substitute f^2 into the equation:

$$ds_{time}^2 = -\left(1 - \frac{r_s}{r}\right)c^2\left(\frac{d\phi^2}{4\pi^2 f_0^2\left(1-\frac{r_s}{r}\right)}\right)$$

Hence, $\left(1 - \frac{r_s}{r}\right)$ terms: **They cancel out completely.**

$$ds_{time}^2 = -\frac{c^2}{4\pi^2 f_0^2}d\phi^2$$

Mathematical Evaluation & Inductive Calculus (Section 4 continuation)

From the fundamental relation governing phase evolution:

$d\phi/dt = 2\pi f \Longrightarrow dt = d\phi\ /\ (2\pi f)$

Evaluating the integral under absolute static conditions ($f \to 0, d\phi \to 0$) yields:

$$\int_{f\to 0} dt = \int_{f\to 0} \frac{d\phi}{2\pi f}$$

If $d\phi = 0$ and $f = 0$, the ratio $\frac{d\phi}{2\pi f}$ yields an indeterminate expression $\left[\frac{0}{0}\right]$

In standard gravitational physics, $\frac{d\phi}{dt} = 0$ is interpreted to be a constant phase ($\phi = C$) over an independent, active background time coordinate ($dt \to \infty$). However, because phase

accumulation and frequency drop to zero simultaneously, the temporal differential dt loses its physical anchor. Time does not stretch infinitely toward an asymptotic limit; rather, the underlying manifold ceases to support a temporal coordinate.

Quantum Constraints and Global Vacuum Preservation (Section 5)

Quantum mechanics states that even within an absolute vacuum or at the absolute zero temperature, quantum fields possess Zero-Point Energy (ZPE) governed by the Heisenberg uncertainty relation:

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

Consequently, suppression of frequency ($f = 0$) across a physical vacuum field is fundamentally forbidden. While local phase-lock nullification can occur under extreme conditions (such as gravitational event horizons), the global vacuum field sustains a non-zero baseline fluctuation rate. This guarantees that phase evolution never entirely ceases on a cosmic scale ($d\phi \neq 0$ globally), preserving the continuous propagation of the temporal metric across open spacetime.

We can formalise this phase evolution by defining a local phase-evolution operator $\hat{U}(\theta) = e^{\left(-i\theta\hat{N}\right)}$, where $\hat{N}$ represents the baseline frequency-state number operator. The infinitesimal generator of temporal displacement is governed by the commutator relation:

$$\frac{d\hat{\phi}}{dt} = \frac{i}{\hbar}[\hat{H}, \hat{\phi}] = 2\pi\hat{f}$$

Evaluating the expectation value over an open thermodynamic system where environmental decoherence and energy dissipation alter the baseline frequency yields:

$$\langle\hat{f}\rangle \to 0 \Rightarrow \langle d\hat{\phi}\rangle \to 0$$

Under these conditions, the temporal expectation metric $\langle dt\rangle$ evaluated via the parameter Fisher information limit collapses to zero variance, confirming that absolute static equilibrium ($f = 0, d\phi = 0$) results in a vanishing Fisher information matrix for temporal duration. The manifold ceases to harbour an active phase-state progression, rendering the time coordinate physically undefined rather than infinitely dilated.

Full Tensor Formulation (Section 6)

Defining the Action Integral

Instead of treating time as an external grid, by introducing a phase scalar field $\phi(x^{\mu})$ where the gradient drives the local temporal metric. By writing the total action S as the sum of the standard Einstein-Hilbert gravitational term and a phase-field kinetic term:

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} - \frac{1}{2}g^{\mu\nu}\partial_{\mu}\phi\partial_{\nu}\phi - V(\phi, f)\right]$$

Where R is the Ricci scalar, g is the determinant of the metric tensor $g_{\mu\nu}$, and $V(\phi, f)$ is the potential energy governing the local frequency baseline $f = \frac{1}{2\pi}\sqrt{(\partial\phi)^2}$.

By varying the action with respect to the metric tensor $g_{\mu\nu}$ yields the modified Einstein field equations:

$$G_{\mu\nu} = 8\pi G T^{(\phi)}_{\mu\nu}$$

Where $T^{(\phi)}_{\mu\nu}$ is the stress-energy tensor generated by the phase field:

$$T^{(\phi)}_{\mu\nu} = \partial_{\mu}\phi\partial_{\nu}\phi - \frac{1}{2}g_{\mu\nu}g^{\alpha\beta}\partial_{\alpha}\phi\partial_{\beta}\phi - g_{\mu\nu}V(\phi, f)$$

Moving beyond static metric substitution, by formulating this behaviour covariantly through a phase-field action $S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} - \frac{1}{2}g^{\mu\nu}\partial_{\mu}\phi\partial_{\nu}\phi\right]$. And by varying this action yields a dynamic stress-energy tensor $T^{(\phi)}_{\mu\nu}$, where local baseline frequency stillness acts as a physical backreaction on the local geometry. Rather than treating the event horizon as an unyielding vacuum singularity, the coupled field equations indicates that the effective spacetime metric self-regulates at $r = r_s$, transitioning the horizon into a finite phase-lock boundary governed by the gradient energy of the phase field.

Discussion and Limitations (Section 7)

Although the wave-mechanics reduction of time resolves the event horizon singularity as $\left[\frac{0}{0}\right]$ a coordinate breakdown, there are several limitations to note:

1. **Empirical Observability:** as the local phase-lock nullification occurs under extreme gravitational conditions only, which is shielded from external observation, direct empirical measurement of localised frequency baseline change rate collapse remains unachievable with current global instrumentation.

SECTION 8:

Spacetime Coordinates & Metric Symbols

x^{μ} : Spacetime coordinates ($\mu = 0, 1, 2, 3$), representing spatial and temporal dimensions in a four-dimensional manifold.

s, ds : Spacetime invariant interval (ds^2) and its infinitesimal element, measuring proper spacetime separation between events.

t, dt : Background temporal coordinate (t) and proper time differential interval (dt).

r : Radial distance coordinate from the centre of mass in spherical geometry.

r_s : Schwarzschild radius ($r_s = \frac{2GM}{c^2}$), defining the gravitational event horizon threshold.

$g_{\mu\nu}$: Spacetime metric tensor, defining the geometric distance properties of the manifold.

g : Determinant of the metric tensor $g_{\mu\nu}$.

c : Speed of light in a vacuum, acting as the fundamental relativistic invariant scale.

Wave Mechanics & Phase Parameters

$\phi(x^{\mu})$: Phase scalar field whose spatial and temporal gradients drive the local temporal metric.

f : Local cyclic frequency baseline of a wave system (Hertz, s^{-1}).

f_0 : Initial or asymptotic background frequency baseline far from gravitational potentials.

A : Wave amplitude parameter.

λ : Spatial wavelength of the wave oscillation.

k : Constant frequency scalar value ($f = k$) representing steady-state cycling.

n : Sequential wave cycle count index ($n = n + 1$).

$d\phi$: Infinitesimal phase accumulation step or phase differential.

Lagrangian Field Theory & Action Symbols

S : Total action integral governing the dynamics of the coupled gravitational and phase-field system.

R : Ricci scalar curvature, representing the local intrinsic curvature of spacetime.

G : Newton's universal gravitational constant.

$V(\phi, f)$: Potential energy function governing the local frequency baseline and scalar field interactions.

$G_{\mu\nu}$: Einstein tensor ($R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$), describing the geometric curvature side of Einstein's field equations.

$T^{(\phi)}_{\mu\nu}$: Phase-field stress-energy tensor, representing the energy-momentum distribution generated by the scalar phase field.

$\partial_{\mu}, \nabla_{\mu}$: Partial derivative (∂_{μ}) and covariant derivative (∇_{μ}) with respect to spacetime coordinate x^{μ}.

Quantum Mechanics & Information Geometry Symbols

$\hat{U}(\theta)$: Local phase-evolution unitary operator ($\hat{U}(\theta) = e^{-i\hat{N}\theta}$).

$\hat{N}$: Baseline frequency-state number operator.

$\hat{H}$: Hamiltonian operator, representing total system energy.

$\hat{\phi}$: Quantum phase operator (the operator counterpart to classical phase ϕ).

$\hbar$: Reduced Planck constant ($\hbar = \frac{h}{2\pi}$).

$\langle \ldots \rangle$: Quantum expectation value / statistical ensemble average over a physical state.

$\hat{f}$: Frequency operator (the operator counterpart to classical frequency f).

Fisher Information Matrix: Information-geometric metric quantifying the sensitivity of a quantum or statistical state to parameter changes (such as temporal duration).

Equations Reference

1. The Dependent Phase-Rate Ratio:
   $dt = \frac{d\phi}{2\pi f}$

   (Defines proper time as a dependent ratio of phase accumulation to local frequency).

2. The Commutator Temporal Generator:

$$\frac{d\hat{\phi}}{dt} = \frac{i}{\hbar}[\hat{H}, \hat{\phi}] = 2\pi\hat{f}$$

(Links the infinitesimal temporal displacement operator to the Hamiltonian-phase commutator and cyclic frequency).

3. Gravitational Redshift Frequency Scaling:

$$f(r) = f_0\sqrt{1 - \frac{r_s}{r}}$$

(Defines how local wave frequency scales with the Schwarzschild metric lapse function).

4. The Transformed Schwarzschild Temporal Metric:

$$ds^2_{time} = -\left(1 - \frac{r_s}{r}\right)c^2\left(\frac{d\phi}{2\pi f}\right)^2$$

(Substitutes the wave-mechanics phase ratio into the metric time component).

5. The Total Field Action Integral:

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} - \frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi, f)\right]$$

(Combines the Einstein-Hilbert gravitational action with the kinetic and potential terms of the phase scalar field).

6. The Modified Einstein Field Equations:

$$G_{\mu\nu} = 8\pi G T^{(\phi)}_{\mu\nu}$$

(Relates spacetime curvature directly to the phase-field stress-energy tensor).

7. The Phase-Field Stress-Energy Tensor:

$$T^{(\phi)}_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi - \frac{1}{2}g_{\mu\nu}g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi - g_{\mu\nu}V(\phi, f)$$ *(Defines the exact energy-momentum tensor derived from varying the phase-field action).*

8. Quantum Vacuum Uncertainty Constraint:

$$\Delta E\Delta t \geq \frac{\hbar}{2}$$

(Heisenberg uncertainty relation enforcing that absolute energy/frequency suppression to zero across a vacuum field is fundamentally forbidden).

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